Program: **Diploma in Microelectronics**	
Course Code: **3498**	Course Title: **Introduction to IoT Lab**
Semester: **3**	Credits: **0**
Course Category: **Program core course**	
Periods per week: **3 (L:0 T:0 P:3)**	Periods per semester: **45**

Course Objectives:

- Provide skills to build Internet of Things (IoT) systems

Course Prerequisites:

Topic	Course code	Course name	Semester
Basics of programming		Electronics Tinkering Workshop	2
Basic Electronics		Basic Electronics	2

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Understand internet of Things and Interface I/O devices, sensors & communication modules	13	Applying
CO2	Remotely monitor data and control devices	9	Applying
CO3	Develop IoT applications using Raspberry Pi	12	Applying
CO4	Design and develop an IoT application to solve problems	8	Applying
	Lab Exam	3	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3			3			
CO2		3		3			
CO3			3	3	2		
CO4			2	3			
CO5	3	3	3	3	3	3	3

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	**Understand internet of Things and Interface I/O devices, sensors & communication modules**		
M1.01	Familiarization of hardware and software components	2	Applying
M1.02	Installation of Arduino Software	2	Applying
M1.03	Controlling the Light Emitting Diode (LED) with a push button	3	Applying
M1.04	Interfacing the RGB LED with the Arduino	3	Applying
M1.05	Controlling the LED blink rate with the potentiometer interfacing with Arduino	3	Applying
	Lab Exam	1.5	
CO2	**Remotely monitor data and control devices**		
M2.01	Detection of the light using photo resistor	3	Applying
M2.02	Interfacing of temperature sensor LM35 with Arduino	3	Applying
M2.03	Interfacing Servo Motor with the Arduino	3	Applying
CO3	**Develop IoT applications using Raspberry Pi**		
M3.01	Experiment with Raspberry PI	4	Applying

M3.02	Develop Simple IoT applications with Raspberry PI to interact with web, mobile applications and cloud	8	Applying
CO4	**Design and develop an IoT application to solve problems**		
M4.01	Design and develop an IoT application to solve a real world problem.	8	Applying
	Lab Exam	1.5	

Text /Reference:

T/R	Book Title/Author
T1	Vijay Madisetti and Arshdeep Bahga, "Internet of Things (A Hands-onApproach)", 1st Edition, VPT, 2016
R2	Richard Blum, "Arduino Programming in 24 Hours", Sams Teach Yourself,Pearson Education,2017
R3	Dr. SRN Reddy, Rachit Thukral and Manasi Mishra, "Introduction to Internet of Things: A practical Approach", ETI Labs
R4	Raj Kamal, Internet of Things : Architecture And Design Principles, McGraw Hill Education

Online Resources:

Sl.No	Website Link
1	https://www.arduino.cc/reference/en
2	https://create.arduino.cc/projecthub
3	https://www.raspberrypi.com/news/getting-started-with-iot/